

Topological Primality — Categorical Extensions (Formal Canvas)

0. Goal

Formalize “Topological Primality” by encoding prime-factor structure of $\mathbb{N}^+$ as categorical/topos-theoretic geometry. The guiding principle is:

Arithmetic locality = prime-factor constraints.

We replace point-set topology on integers with a **site** built from divisibility, then define sheaves, (co)homology-like invariants, and categorical “retractions” expressing prime-set structure.

1. Divisibility category (the base site)

- **Objects:** positive integers $n \in \mathbb{N}^+$.
- **Morphisms:** a unique arrow $m \rightarrow n$ iff $m \mid n$.

Equivalently, **Div** is the poset $(\mathbb{N}^+, \mid)$ viewed as a category.

1.2 Principal up-sets

For $n \in \mathbb{N}^+$, define the principal upset

$$\uparrow n := \{m \in \mathbb{N}^+ : n \mid m\}.$$

These are the categorical avatars of “basic opens”: they encode all integers satisfying at least the prime-valuation constraints of n .

2. Prime-valuation geometry

2.1 Valuation vector embedding

Let $v_p(n)$ denote the exponent of prime p in n . Define

$$\nu: \mathbb{N}^+ \rightarrow \bigoplus_{p \in \mathbb{P}} \mathbb{N}, \quad \nu(n) = (v_p(n))_p.$$

Then

$$m \mid n \text{ iff } v_p(m) \leq v_p(n) \text{ for all } p.$$

Thus **Div** is equivalent to the finite-support product poset $\bigoplus_p \mathbb{N}$ with coordinatewise order.

2.2 Prime-support regions (multiplicity-aware)

For finite $S \subset \mathbb{P}$ and multi-index $\mathbf{k} = (k_p)_{p \in S} \in \mathbb{N}^S$, define

$$U_S^{\mathbf{k}} := \{n \in \mathbb{N}^+ : \forall p \in S, v_p(n) \geq k_p\}.$$

Each $U_S^{\mathbf{k}}$ is an upset in $(\mathbb{N}^+, |)$, hence corresponds to a sieve on **Div**.

3. Grothendieck topologies (prime local generation)

3.1 Coverage by unions of principal up-sets

A family $\{n_i \rightarrow n\}_i$ is a **cover** of n if

$$n \leq \bigcup_i n_i.$$

Equivalently: for every m with $n \mid m$, there exists i such that $n_i \mid m$.

This defines a Grothendieck topology $J_{\cup}$ on **Div** (posetal “union coverage”).

3.2 Prime-generated refinement (optional)

To enforce explicit prime locality, restrict to covers where each n_i refines n by controlled prime-power increments, e.g.

$$n_i = n \cdot \prod_{p \in S_i} p^{t_{i,p}},$$

for finite S_i and exponents $t_{i,p} \geq 0$. Call the resulting topology J_{prime} .

4. Sheaves and “prime cohomology”

4.1 Presheaves

A presheaf of sets/abelian groups is a functor

$$\mathcal{F} : \mathbf{Div}^{\text{op}} \rightarrow \mathbf{Set} \text{ or } \mathbf{Ab}.$$

- $\mathcal{F}(n)$ is the data visible at “resolution” n .
- If $m \mid n$, the unique arrow $m \rightarrow n$ induces a restriction $\mathcal{F}(n) \rightarrow \mathcal{F}(m)$.

4.2 Sheaves

A presheaf $\mathcal{F}$ is a **sheaf** (for J_{prime}) if for every covering family $\{n_i \rightarrow n\}$:

- sections over n are determined uniquely by compatible sections over each n_i ,
- with compatibility enforced on overlaps (via pullbacks in the poset, typically lcms).

4.3 Global sections and derived invariants

Global sections are

$$\Gamma(\mathcal{F}) = \mathcal{F}(1),$$

since $1 \mid n$ for all n so 1 is terminal in $\mathbf{Div}$.

Define “prime cohomology” as derived functors in the topos

$$H^i(\mathbf{Div}, J_{\text{prime}}; \mathcal{F}) = R^i \Gamma(\mathcal{F}).$$

Note: Because $\mathbf{Div}$ is filtered, many constant-coefficient cohomologies may vanish in degrees > 0 . Prime-sensitive variants below address this.

5. Prime-set deformation retracts as adjunctions/(co)monads

5.1 Squarefree coreflection via radical

Define the radical (squarefree part)

$$\text{rad}(n) = \prod_{p \mid n} p.$$

Let $\mathbf{Sqf} \subset \mathbf{Div}$ be the full subcategory of squarefree integers.

There is an inclusion $i : \mathbf{Sqf} \hookrightarrow \mathbf{Div}$ and a functor

$$\text{rad} : \mathbf{Div} \rightarrow \mathbf{Sqf}$$

with $\text{rad}(n) \mid n$. This yields an idempotent closure/interior operator on the poset, categorically expressible as an idempotent (co)monad (variance depending).

Interpretation: “retracting” integers to their prime-sets is better behaved via $\mathbf{Sqf}$ than via $\mathbb{P}$ alone.

5.2 Support functor to finite prime sets

Define

$$\mathrm{Supp}(n) = \{p \in \mathbb{P} : p \mid n\} \in \mathbf{FinSet}.$$

Then $\mathrm{Supp}(mn) = \mathrm{Supp}(m) \cup \mathrm{Supp}(n)$. This is a monoid map into $(\mathcal{P}_{\mathrm{fin}}(\mathbb{P}), \cup)$.

6. Non-Archimedean enrichment (optional)

6.1 Support-distance and multiplicity-distance

Examples of prime-sensitive pseudometrics:

- Support overlap metric

$$d_{\mathrm{cap}}(m, n) = \frac{1}{1 + |\mathrm{Supp}(m) \cap \mathrm{Supp}(n)|}.$$

- Multiplicity ultrametric-like form (heuristic)

$$d_{\mathrm{nu}}(m, n) = \prod_{p \in \mathbb{P}} p^{-|v_p(m) - v_p(n)|}.$$

These can be used to enrich **Div** or to define enriched sheaves / continuity constraints.

7. Avoiding triviality: prime-sensitive invariants

7.1 Truncated sites

Define subsites to prevent global filtering from collapsing invariants:

- Prime cutoff: $\mathbf{Div}^{(\leq N)}$: objects with primes $\leq N$.
- Valuation bound: $\mathbf{Div}^{(\leq K)}$: objects with $\sum_p v_p(n) \leq K$.
- Fixed support: $\mathbf{Div}_S$: primes restricted to a finite set S .

Study cohomology on these finite/controlled sites and then pass to limits as $N \rightarrow \infty$, etc.

7.2 Equivariant / action-based constructions

Let $(\mathbb{N}^+, \cdot)$ act on **Div** by endofunctors

$$T_k(n) = kn.$$

This induces an action on $\mathbf{Sh}(\mathbf{Div}, J_{\mathrm{prime}})$. Prime-sensitive invariants may be defined via fixed points or derived limits, e.g.

$$\Gamma(\mathcal{F})^{\mathbb{N}^+} \simeq \mathrm{holim}_k \Gamma(T_k^* \mathcal{F}).$$

8. Predictions / expected outcomes

1. **Squarefree reduction is canonical:** $\mathbf{Sqf}$ behaves as a coreflective “prime-set skeleton,” allowing functorial projection rad without arbitrary choices.
 2. **Truncation reveals structure:** cohomological invariants on $\mathbf{Div}^{(\leq N)}$ should show growth patterns reflecting prime distribution in $[2, N]$.
 3. **Equivariant invariants detect primes:** invariants built from the action of T_p (prime multiplication) should separate prime-power strata from mixed composites.
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9. Fast validation program (minimal)

1. **Choose a Grothendieck topology** $J_{\cup}$ or J_{prime} and characterize sheaves via gluing along lcms.
 2. **Compute cohomology** for a small library of coefficient sheaves on $\mathbf{Div}^{(\leq N)}$ for $N = 2, 3, 5, 7$.
 3. **Test filtered-triviality:** check whether constant sheaves have vanishing higher cohomology; if so, pivot to truncated/equivariant invariants.
 4. **Implement toy computations:** build the Čech complex for covers generated by prime-power refinements and measure nontriviality.
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10. Next formal tasks

- (A) Pick a canonical prime-generated Grothendieck topology J_{prime} and prove the Grothendieck axioms.
 - (B) Describe points of the topos $\text{Sh}(\text{Div}, J_{\text{prime}})$ (i.e., geometric morphisms from Set).
 - (C) Define a prime-sensitive coefficient system F (e.g., valuation-weighted or Möbius-twisted) and compute low-degree invariants.
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11. References (selected)

Topos theory, sites, and sheaves

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Non-Archimedean / ultrametric background

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- W. H. Schikhof. *Ultrametric Calculus: An Introduction to p-adic Analysis*, Cambridge Studies in Advanced Mathematics 4, Cambridge University Press, 1984.

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Classical scheme-theoretic background (comparison target)

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Absolute / F_1 / monoid-scheme context (optional comparison)

- A. Deitmar. "Schemes over F_1 ," arXiv:math/0404185 (versions through 2005).